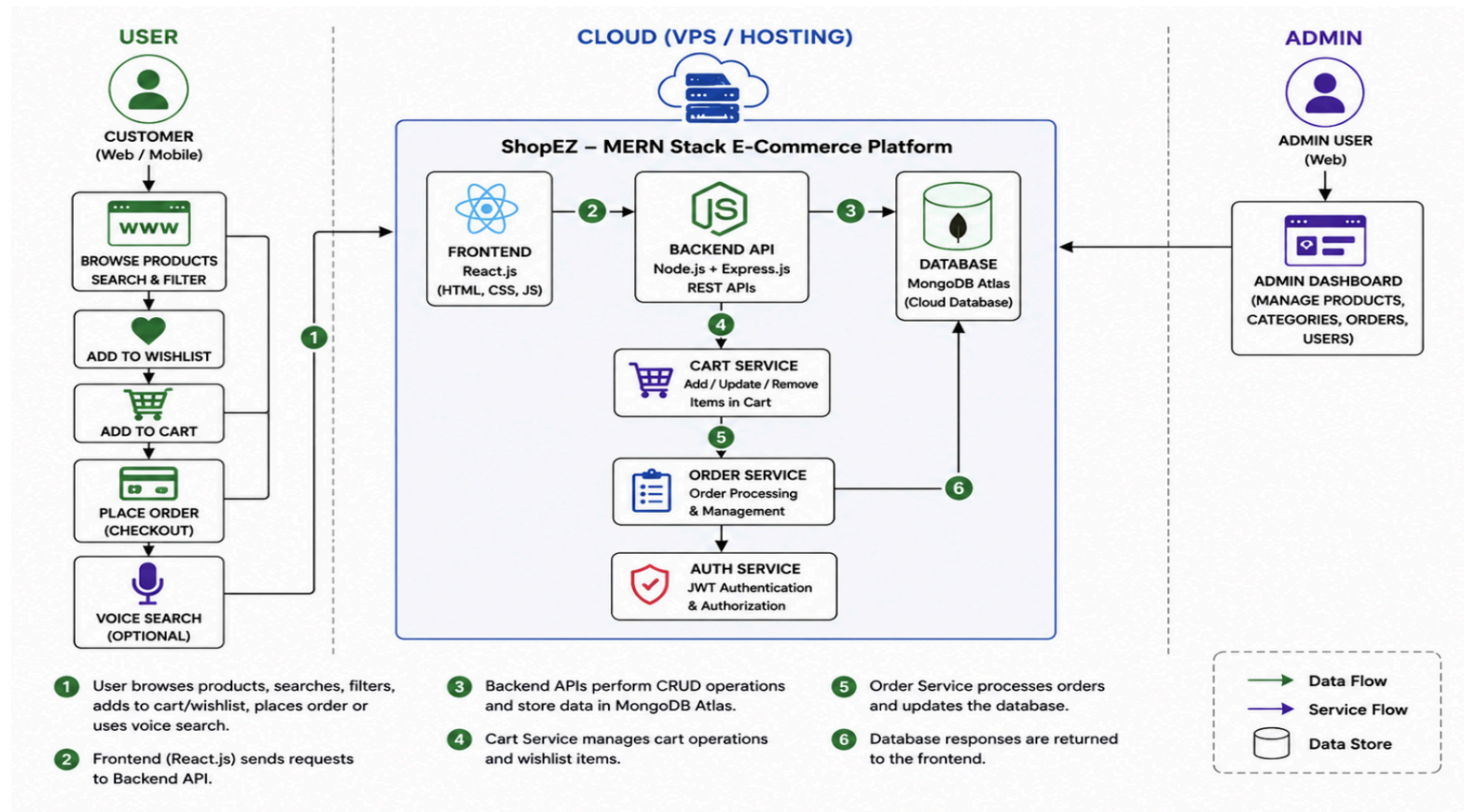


## Project Design Phase-II Technology Stack (Architecture & Stack)

|               |                                         |
|---------------|-----------------------------------------|
| Date          | 18 June 2026                            |
| Team ID       |                                         |
| Project Name  | ShopEZ – MERN Stack E-Commerce Platform |
| Maximum Marks | 4 Marks                                 |

### Technical Architecture:



**Table-1 : Components & Technologies:**

| S.No | Component                       | Description                                                 | Technology                      |
|------|---------------------------------|-------------------------------------------------------------|---------------------------------|
| 1.   | User Interface                  | Web application for customers and administrators            | React.js, HTML, CSS, JavaScript |
| 2.   | Application Logic-1             | User registration, login, authentication, and authorization | Node.js, Express.js, JWT        |
| 3.   | Application Logic-2             | Product listing, search, filtering, and wishlist management | Express.js REST APIs            |
| 4.   | Application Logic-3             | Shopping cart, quantity management, and order processing    | Node.js, Express.js             |
| 5.   | Database                        | Stores users, products, wishlist, cart items, and orders    | MongoDB                         |
| 6.   | Cloud Database                  | Database hosting and management                             | MongoDB Atlas                   |
| 7.   | File Storage                    | Product image URLs and image references                     | Cloud Image URLs                |
| 8.   | External API-1                  | HTTP requests between frontend and backend                  | Axios                           |
| 9.   | External API-2                  | Authentication and authorization tokens                     | JWT                             |
| 10.  | Machine Learning Model          | Not Applicable                                              | N/A                             |
| 11.  | Infrastructure (Server / Cloud) | MERN Stack deployment environment                           | Local Server / Cloud Deployment |

**Table-2: Application Characteristics:**

| S.No | Characteristics          | Description                                                     | Technology                             |
|------|--------------------------|-----------------------------------------------------------------|----------------------------------------|
| 1    | Open-Source Frameworks   | Frontend and backend frameworks used in ShopEZ                  | React.js, Node.js, Express.js, MongoDB |
| 2    | Security Implementations | User authentication, authorization, and protected admin routes  | JWT Authentication, Password Hashing   |
| 3    | Scalable Architecture    | Three-tier MERN architecture supporting future growth           | MERN Stack Architecture                |
| 4    | Availability             | Accessible through web browsers anytime and anywhere            | Node.js Server + MongoDB Atlas         |
| 5    | Performance              | Fast product search, filtering, and API responses               | Express.js, MongoDB Indexing           |
| 6    | Reliability              | Consistent access to products, cart, wishlist, and orders       | MongoDB Atlas, REST APIs               |
| 7    | Maintainability          | Modular code structure for easy updates and debugging           | React Components, Express Routes       |
| 8    | Compatibility            | Supports modern browsers and multiple devices                   | Chrome, Firefox, Edge, Safari          |
| 9    | Data Integrity           | Accurate storage and retrieval of user, product, and order data | MongoDB Database                       |